

Transfer without Forgetting

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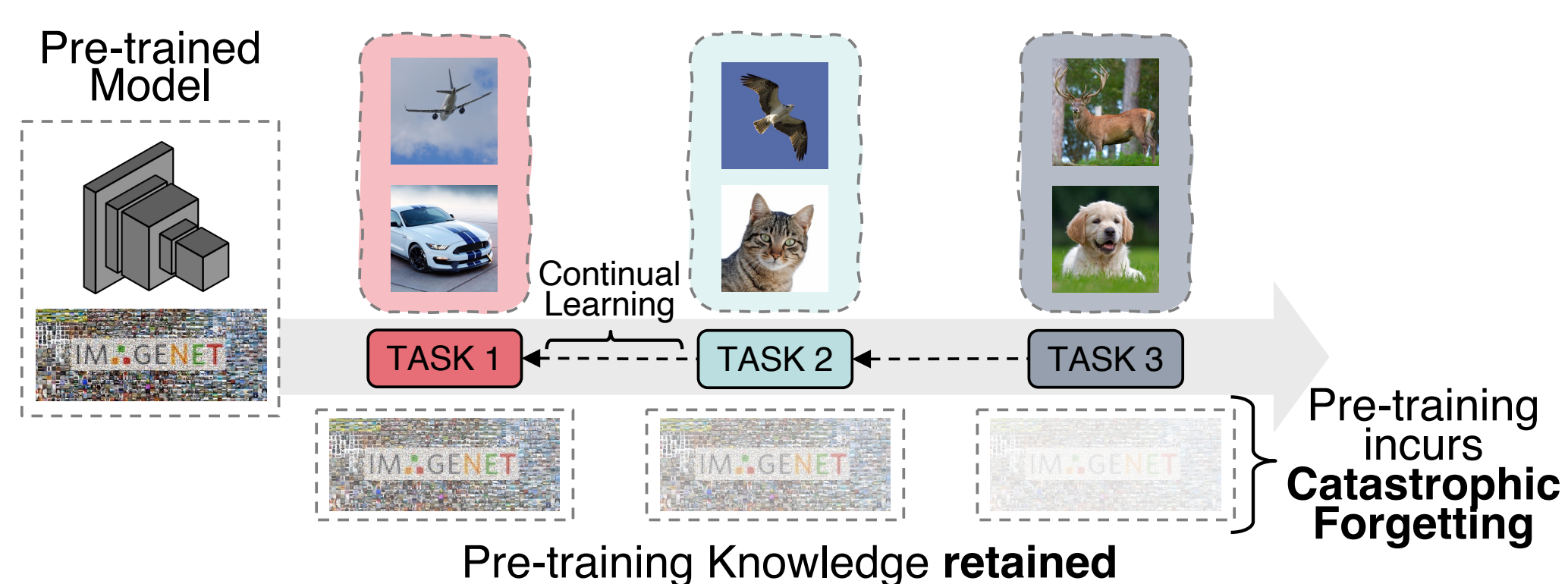
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1. Motivation

Continual Learning (CL) models seek to avoid *catastrophic forgetting* and retain old knowledge in subsequent tasks.

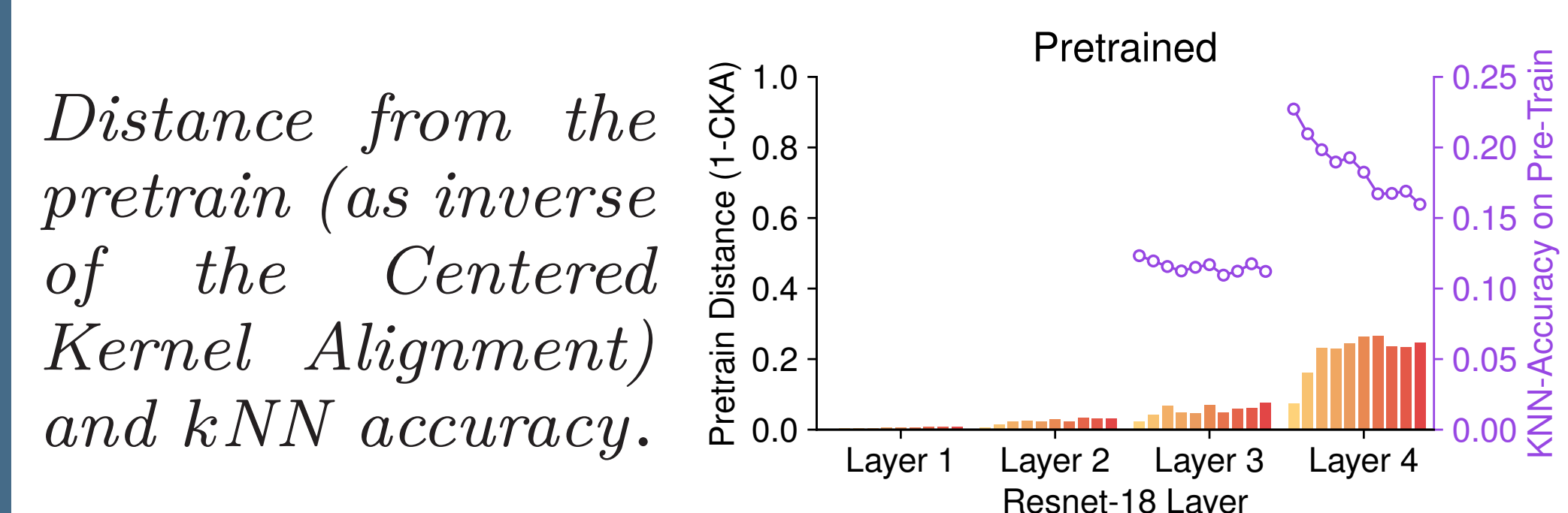
Problem: when *pretraining* is used, it is itself catastrophically forgotten as the model veers towards the newly introduced stream of data.



Consequence: state-of-the-art CL methods leveraging pretraining knowledge **do not take full advantage of it** (as pretraining ends up being fully available only during the former tasks).

2. Pre-training incurs forgetting

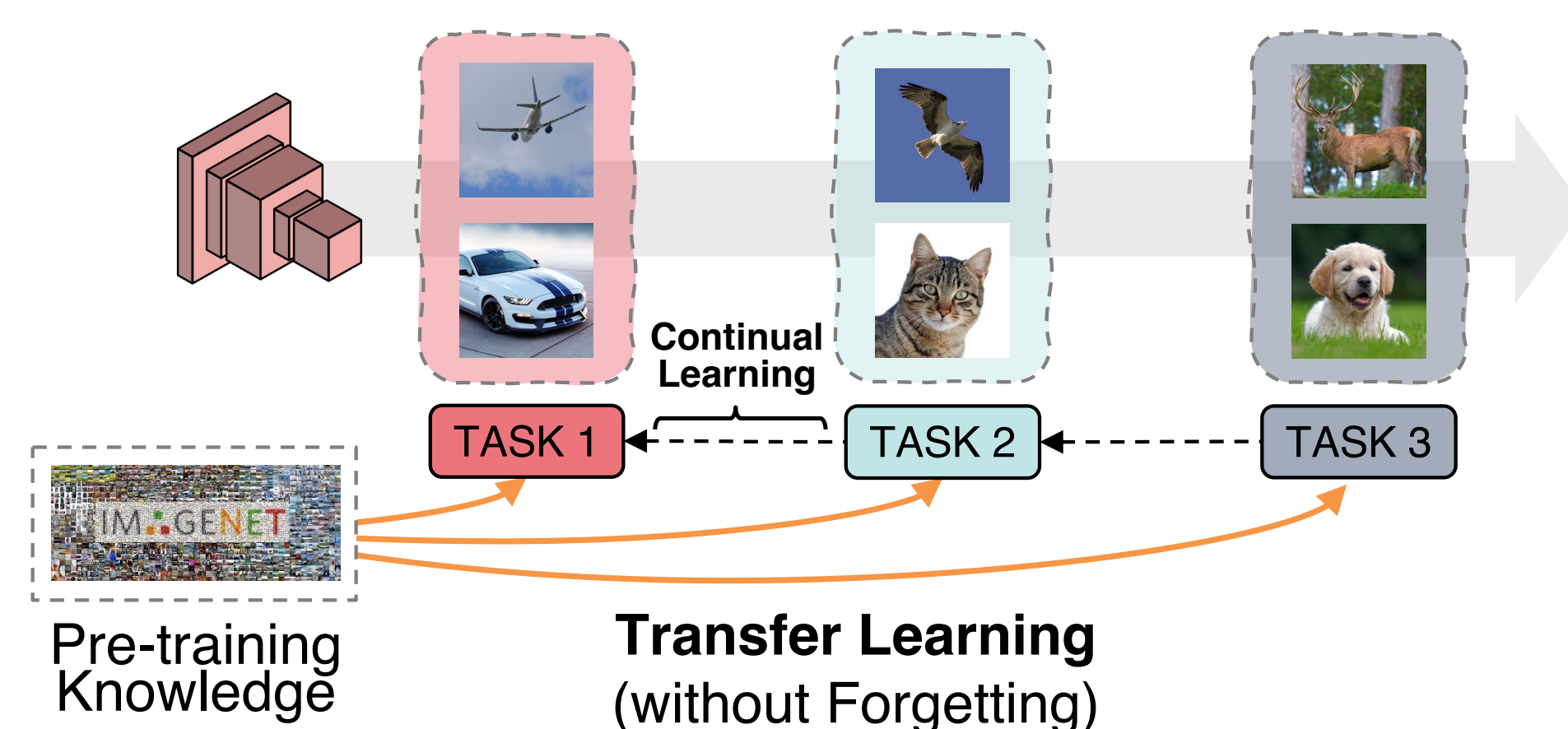
Intuition: the transfer mostly concerns the early encountered classes; for later ones, as pretraining weights underwent changes, the benefit of pretrain reduces.



The subsequent tasks should be likewise advantaged by the (partial) reuse of the pretraining knowledge.

3. Idea

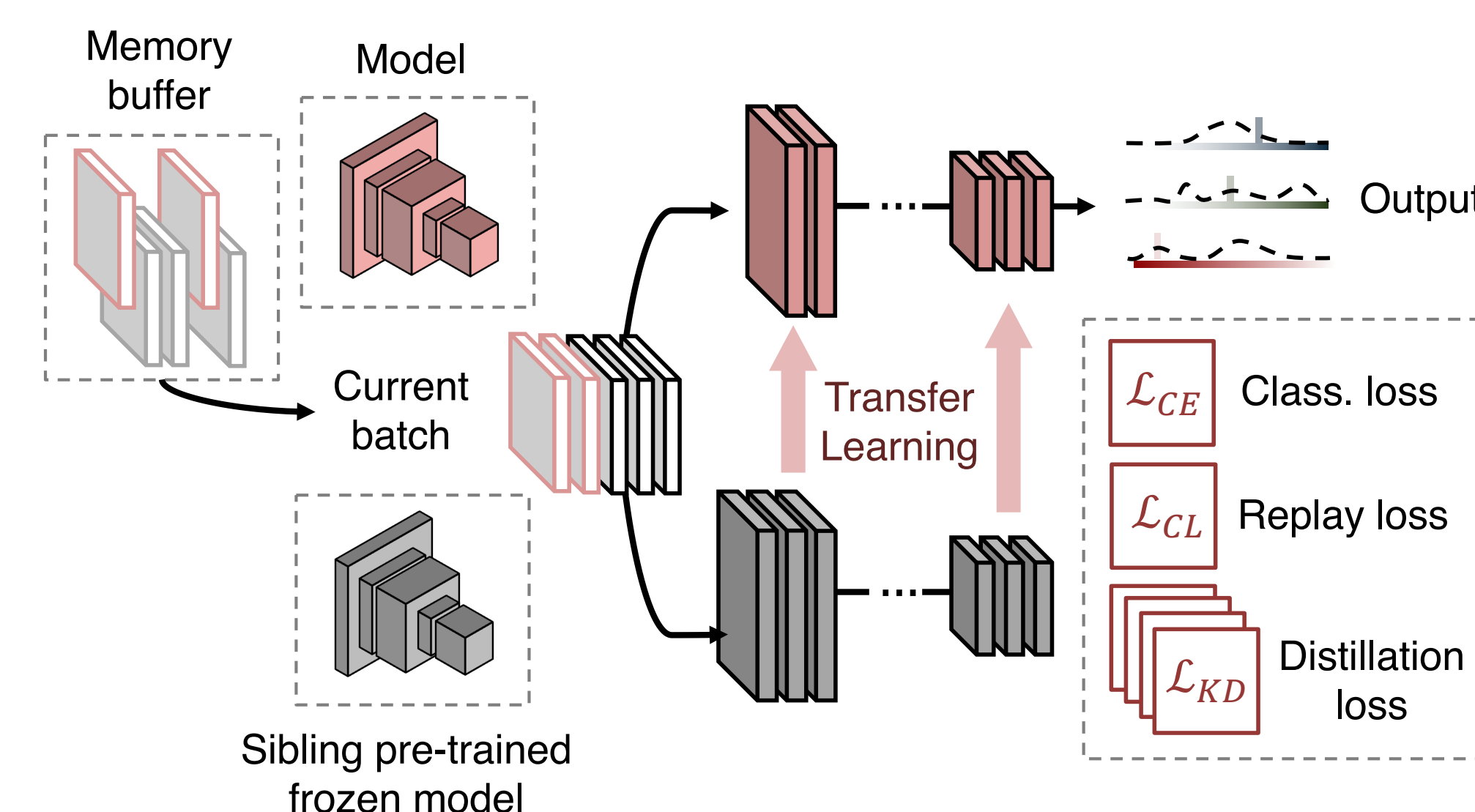
Approach: we argue that the CL setting requires **tailored** strategies to exploit the source knowledge without incurring its forgetting.



To let all tasks profit equally from pretraining, we propose **Transfer Without Forgetting**, which enables a **continuous transfer** between the source task and the incrementally learned target problem.

4. Transfer without Forgetting

Overall: TwF equips a base rehearsal model with a **pretrained** and **fixed** sibling network.



The pretrained sibling network **continuously transfers** its representations through a **per-layer strategy** based on **Knowledge Distillation**.

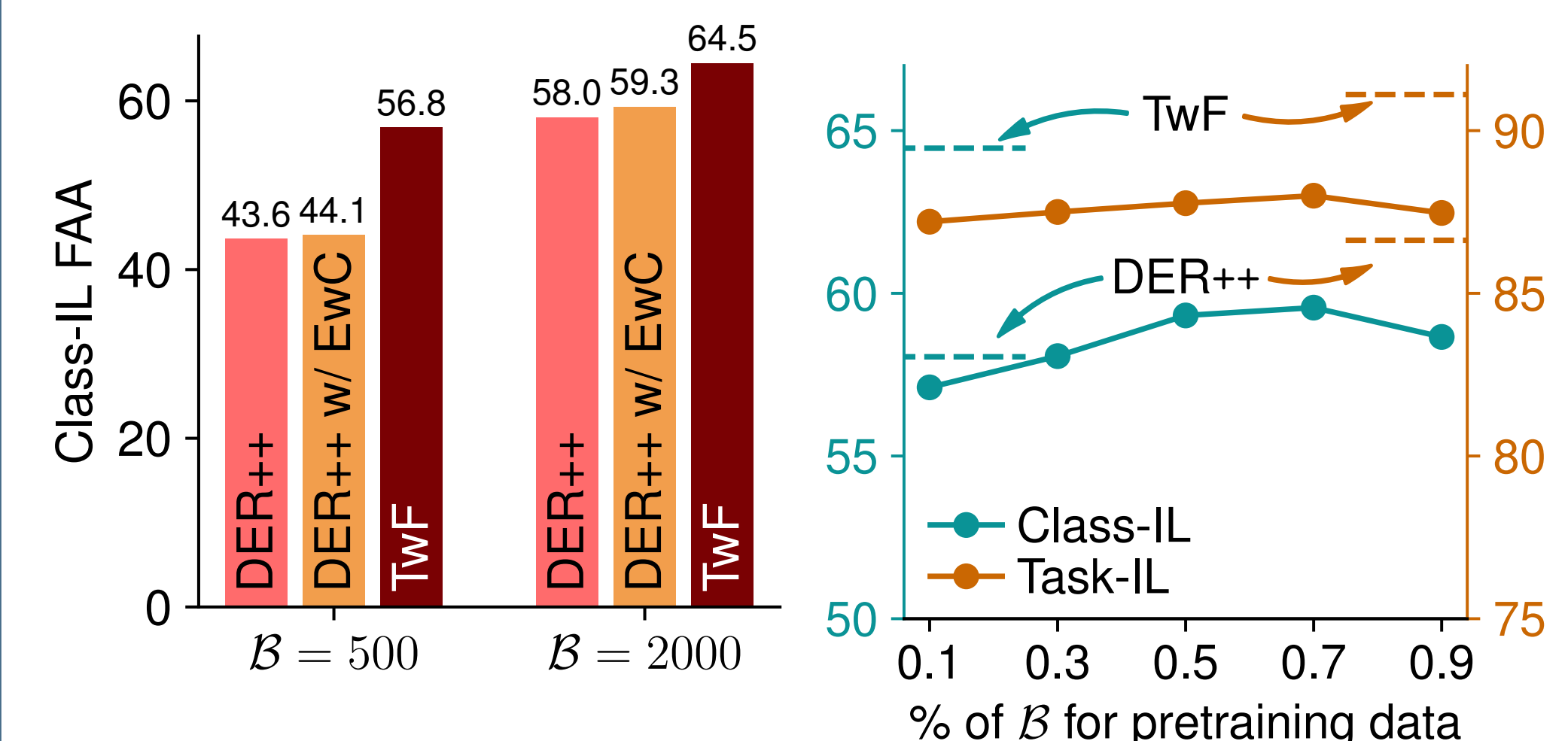
5. Comparison with SOTA

FAA (FF)	Split CIFAR-100		Split CUB-200	
Buffer Size	500	2000	400	1000
ER	31.30 (65.40)	46.80 (46.95)	45.82 (40.76)	59.88 (25.65)
CO ² L	33.40 (45.21)	50.95 (31.20)	8.96 (32.04)	16.53 (20.99)
iCaRL	56.00 (19.27)	58.10 (16.89)	46.55 (12.48)	49.07 (11.24)
DER++	43.65 (48.72)	58.05 (29.65)	56.38 (26.59)	67.35 (13.47)
ER-ACE	53.38 (21.63)	57.73 (17.12)	48.18 (25.79)	58.19 (16.56)
TwF (ours)	56.83 (23.89)	64.46 (15.23)	57.78 (18.32)	68.32 (6.74)

All these approaches start from ImageNet pre-training, but **protecting it against forgetting (as done by TwF)** leads to **better results**.

6. Alternative pre-train defenses

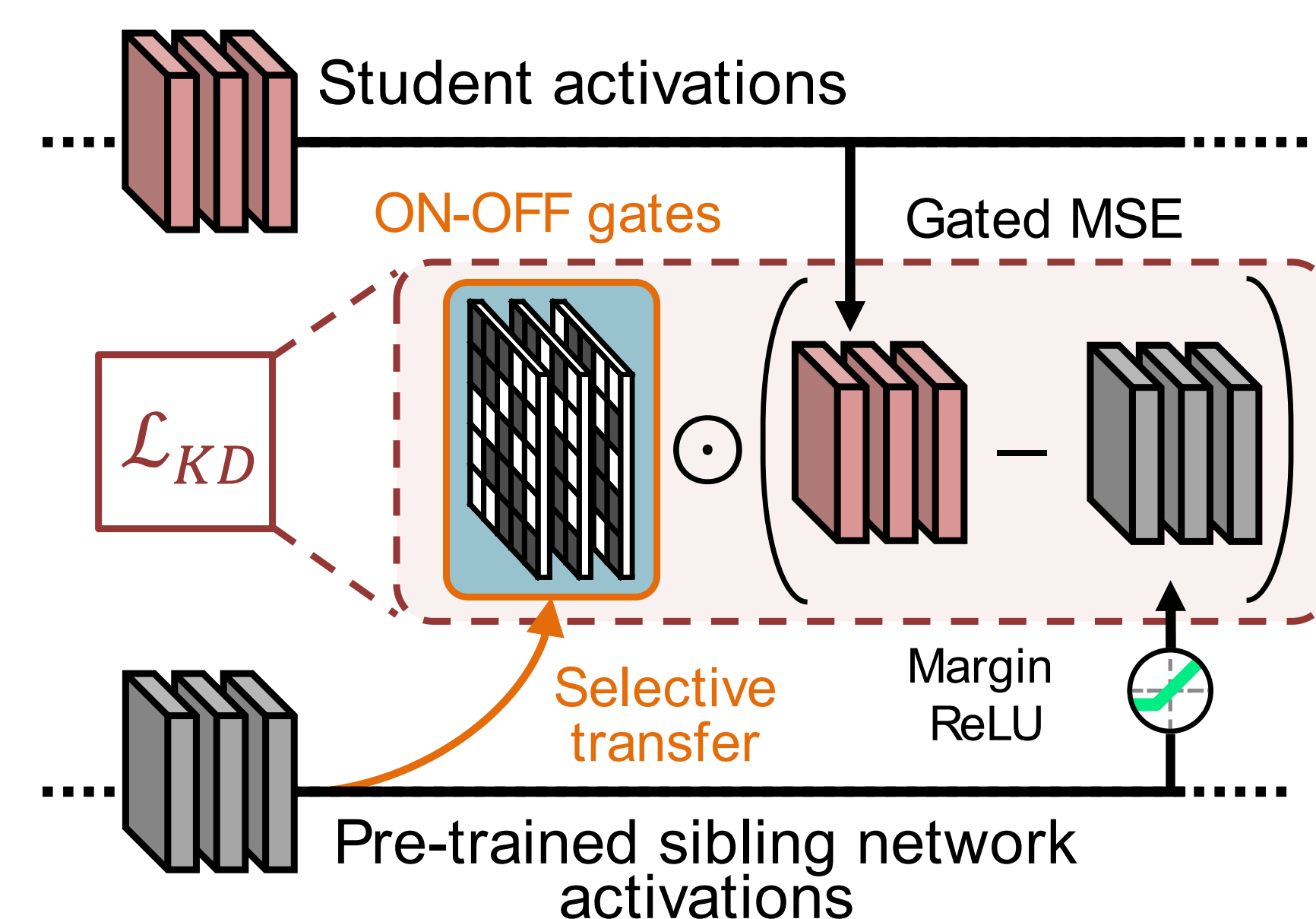
Alternative CL regularization strategies could be considered for preserving pre-train knowledge.



Compared to EwC and ER, TwF provides **better performance**.

4a. Adaptive Transfer

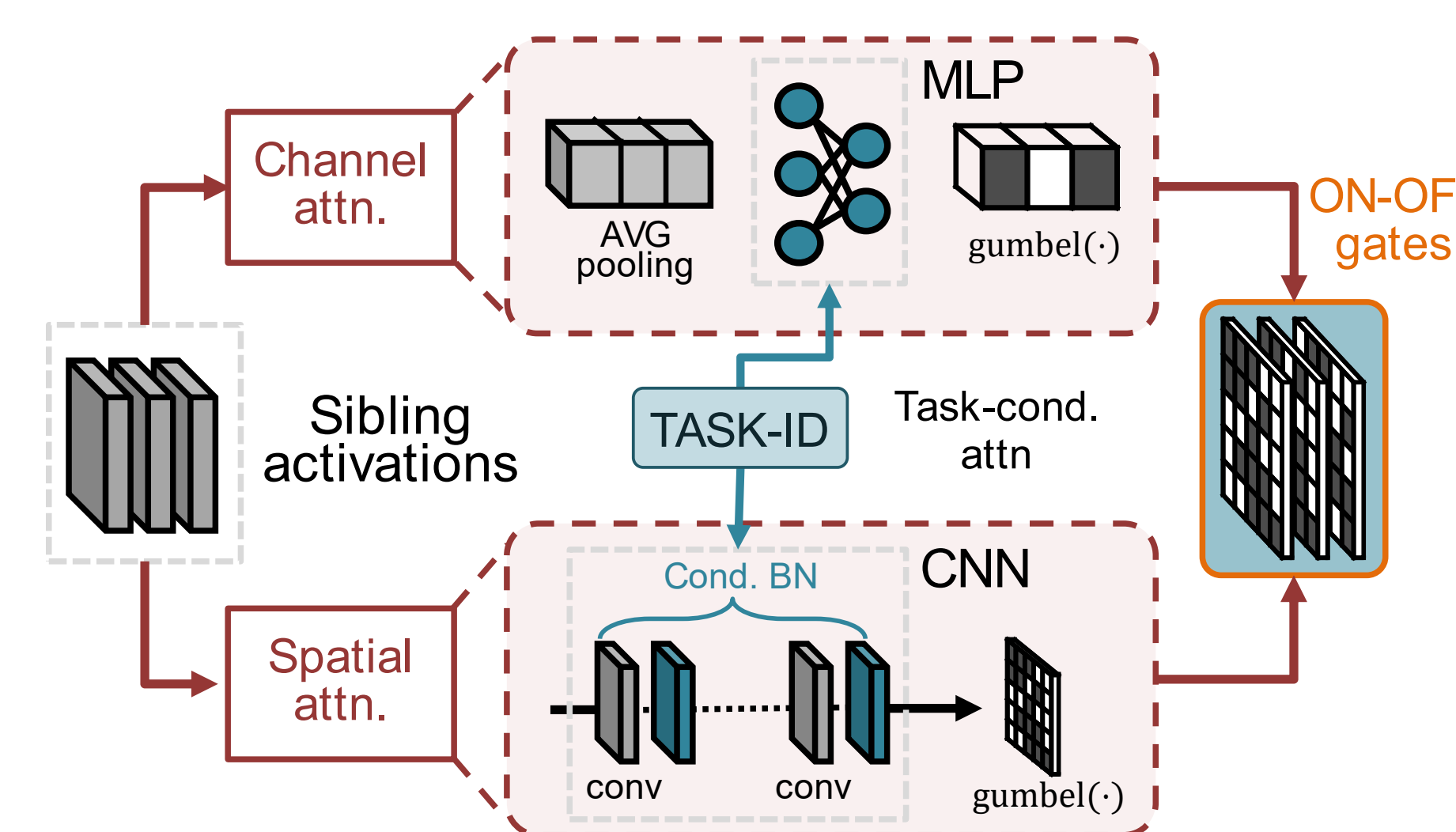
TwF aligns hidden representations by **masking out** the Mean Squared Error adaptively.



The gates are computed by **auxiliary modules** fed with the activations of the sibling network.

4b. Task-dependent Transfer

To reduce cross-task interference, the weighing gates are **conditioned on task-ids** and **stored for later replay** (as for labels and old responses).



Diverse loss: we also encourage the auxiliary modules to assign different gating masks to different examples.

7. Dissimilar pre-train task

SVHN 680 3' 5 → Split CIFAR-100

FAA (FF)	Class-IL		Task-IL	
Buffer size	500	2000	500	2000
iCaRL	39.59 (21.81)	42.02 (18.78)	78.89 (4.04)	80.65 (2.24)
DER++	36.46 (53.47)	52.29 (24.04)	75.05 (16.22)	83.36 (8.04)
TwF (ours)	43.56 (40.02)	56.15 (21.51)	80.89 (10.12)	87.30 (3.12)

Robustness w.r.t. bad pre-train: TwF transfers only the pretraining features that **really help**.